Russell Mercer

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Biochemistry student preparing for medical school, who designs the tools and interfaces clinicians and researchers use, from radiation oncology to the lab bench. Enjoys learning new things and being challenged in new ways.

EDUCATION

Brigham Young University · *Provo, UT* Jan 2024 – Apr 2028 (Expected)
B.S. Biochemistry; Minor in Computer Science, with machine learning courses
GPA 3.99 · Science GPA 3.99 · MCAT 519 (96th percentile)

DESIGN AND BUILD PROJECTS

Clinical Imaging Workspace · *UT Southwestern* 2026

- Designed and built the interface physicians and medical physicists use: viewer with per-day histograms, parameter comparison, publication image creator, threshold creator, and a fit checker answering whether a number is real
- Iterated from physician feedback; the diffusion overlay let a physician redo 46 contours

Figures, Posters, and Identity · *BYU and UT Southwestern* 2025 – 2026

- Designed a treatment-workflow comparison figure assembled automatically from slide art and R panels, so revisions cost nothing
- Six research posters and talk decks across chemistry, medicine, and structural biology
- Rebuilt the SPRI program mark on an axis model after feedback showed the original read wrong; delivered as an interactive kit

RESEARCH EXPERIENCE

Biochemistry Research Assistant · *Jensen Lab, BYU* Aug 2026 – Present

- Two cryo-electron tomography projects I scoped and the PI approved, awaiting biosafety review; AI agent study presented at the Utah CryoEM Symposium 2026

Clinical Research Assistant · *UT Southwestern Medical Center, Dallas, TX* May 2026 – Present

- Four studies of MR-guided and biology-guided radiotherapy; first author on three manuscripts, one submitted and two in final co-author circulation

Chemistry Research Assistant · *Stowers Lab, BYU* Apr 2024 – May 2026

- Second author, *Langmuir* 2026; built and maintained a flow reactor and a UHV chamber (~10^{-10} Torr, XPS); trained two newer students

LEADERSHIP AND SERVICE

Officer · *BYU AI Association* Sep 2026 – Present

- Lead the weekly AI in Research meeting; speak about AI in research to other colleges across the university and meet with professors and labs

Lead Program Director · *BYU Y-Serve, Early Learning Essentials* Aug 2024 – Present

- Run a volunteer program serving preschoolers from low-income families

Pre-Med Peer Mentor · *BYU Pre-Med Association* Oct 2026 – Present

- Mentor pre-med students; co-taught two university courses as a BYU peer mentor, 2024 – 2025

SKILLS

- **Design:** user research with clinicians, interface design, data visualization, figure and poster design, rapid prototyping
- **Tools:** PowerPoint, ImageJ, ggplot2 and matplotlib, HTML and CSS, SVG
- **Code:** Python, R, C#, Java, AI coding agents
- **Lab and Clinical:** SEM, UHV, cryo-EM sample preparation, clinical data abstraction

PUBLICATIONS

- Ralph AD, **Mercer RW**, et al. *Langmuir* 2026;42(6):4475–88.
- **First author on three manuscripts:** one submitted (*Int J Radiat Oncol Biol Phys*) and two in final co-author circulation.
- Co-author on two more in final circulation, with others in preparation.

AWARDS

- **SCCR Fellow**, BYU Simmons Center for Cancer Research, 2026
- **Outstanding Analytical Chemist**, 2025
- **Dean's List**, four semesters, 2024 – 2025
- **Undergraduate Research Award**, each semester, Stowers Lab

PRESENTATIONS

- Utah CryoEM Symposium, 2026
- BYU Simmons Center for Cancer Research, 2026
- BYU Medical Society, 2026
- ACS Spring Meeting, Atlanta, 2026
- BYU Student Research Conference, 2025 and 2026